AI × CYBERSECURITY · SEMESTER PROJECT · MUST UNIVERSITY

# Voice-Spoofing Detection for Fraud Prevention

Execution Plan v2.1: how we build the project, stage by stage, and who owns what


**TEAM**
Donia Mabrouk · Iheb Zemzemi · Malak Ben Salem · Khalil


**REPOSITORY**
github.com/Doniaa00/voice-spoofing-detection

**VERSION / DATE**
v2.1 · 12 October 2026 (replaces v2.0 and the v1 Execution Reference)

**CONTROLLING DOCUMENTS**
Decision register (D1–D25) › SRS v3.0 › SDD v1.0 › this plan › Proposal

## Where we are today

| | |
|---|---|
| DONE Stages 0–2 | Repository, dataset audit (signed off), threat model v1.0 (reviewed); decisions D1–D21 logged. |
| DONE Repo structure | PR #6 merged: CS 488 Lab 6 template adapted to the project (AGENTS.md, workflow, CI, docs). CI is required on `main`. |
| NEW Team of four | Khalil joined on 12 October (approved by the team and the professor). Every part now has an owner (Section 5). |
| NEXT Stage 3 | Experimental Protocol v1.0: fixes the experiment's rules *before* any training code. |

Every fact marked “verified” in this plan comes from committed evidence in the repository ( `docs/audit_evidence/` , `docs/dataset_audit.md` ).

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# 1. How to use this plan

This is the team's working reference for the whole semester. It says **what** we build, **in which order**, and **what "done" means** at each stage. It does not repeat everything: details live in the documents it points to.

- **Start of a stage:** read its card (Sections 7–11) and check its owner (Section 5): goal, tasks, outputs, gate, and the decisions that apply.
- **During a stage:** follow the hard rules (Section 3) and the workflow (Section 4).
- **End of a stage:** the stage is done only when its **gate** is passed: outputs merged into `main` through a reviewed pull request.
- **Something conflicts?** The decision register wins, then SRS, then SDD, then this plan. Never resolve a conflict silently: raise it, propose a fix, decide as a team.

> **Why v2.x?** The v1 plan was a good starting point, but the readiness review and the dataset audit showed several places where following it literally would have weakened the results (for example, testing on DEEP-VOICE before thresholds existed). v2.0 keeps the same goal, phases and deliverables, and adds the corrections the team approved. Appendix D lists every change. v2.1 adds the team of four and the ownership of every stage.

# 2. The project on one page

### What we build

A **decision-support** security control that analyses a short recorded speech segment and returns **authentic / synthetic / uncertain**, a **confidence** score and a **LOW / MEDIUM / HIGH risk level**. It helps a fraud analyst decide whether a high-risk phone request (transfer, account change, credential reset) needs extra verification. It is **not** a biometric authenticator and **not** proof of fraud.

### Research question (D8)

How much does a detector trained on **text-to-speech** fakes (FoR-2sec) lose when it is evaluated **once, frozen**, on **voice-conversion** fakes (DEEP-VOICE, RVC)? The two datasets differ in several ways at once (generator, speakers, recording conditions, format), so we measure *how much* performance changes; attributing the change to one cause stays a hypothesis.

### Datasets (verified in Stage 1)

| Dataset | Role | Verified facts |
|---|---|---|
| **FoR-2sec**<br>HF UncovAI/FOR-2sec, rev. ff8c82c7 | Train, validation, in-domain test | 17,721 clips: 8,921 FAKE / 8,800 REAL. All 16 kHz mono, exactly 2.0 s. 0 corrupt, 0 duplicates; 98 near-duplicate pairs (largest group 6 clips). No packaged split, no speaker metadata. **85% of FAKE clips (0% of REAL) had MP3 history**; FAKE carries less high-frequency energy (≈3.3 kHz vs ≈5.1 kHz). |
| **DEEP-VOICE**<br>Kaggle birdy654, zip SHA-256 8cb25853… | External test, **used exactly once** | 64 files: 8 REAL + 56 FAKE (8 public figures, each converted into the 7 others). Stereo, 40 / 44.1 / 48 kHz, 79–600 s per file. As 2-s windows: 1,870 REAL / 13,090 FAKE (**87.5% FAKE**). 0 files shared with FoR. MIT license. Inspected by metadata only. |

### Scope lock

**In scope**

- Baseline: MFCC (mean + std) + RBF-SVM, evaluation only (D7)
- One deployed model: a CNN on log-Mel spectrograms
- Risk layer: thresholds, uncertain band, calibration
- Two stretch experiments: robustness curve, calibration analysis
- FastAPI service, Streamlit dashboard, Docker

**Out of scope (future work)**

- wav2vec2 / WavLM / HuBERT / AASIST / RawNet, ensembles
- Other features (LFCC, CQCC, raw waveform), explainability
- Live / streaming detection, adversarial defense, replay attacks
- Data augmentation in model v1 (D10)

### Metrics (D4)

Positive class = spoof. **Score level** (threshold-free): EER, ROC-AUC. **Decision level** (at the frozen threshold): accuracy, precision, recall, F1, both forced-binary and on non-deferred cases. Always reported with a **3-way confusion matrix** (incl. uncertain), coverage / deferral rate, class counts, FPR / FNR and 95% confidence intervals. Never accuracy alone. No t-DCF.

## 3. Hard rules

These rules protect the validity of the results. They are written in `AGENTS.md`, so every AI tool we use follows them too. Breaking one can invalidate months of work.

1. **DEEP-VOICE is touched exactly once.** Never used for training, tuning, thresholds, calibration, model selection or exploration; no listening, no plots, no feature statistics. Before the single external run, the only permitted accesses are download, zip hash, unzip, listing file names/sizes/hashes, and the metadata audit script (D20). A CI test enforces this in code.
2. **The ModelBundle is frozen as one unit** (CNN checkpoint + preprocessing profile + risk policy), fitted on FoR only, **before** the external run (D2).
3. **Scope lock:** one baseline, one CNN. No extra models or features (Section 2).
4. **No training code** before Protocol v1.0 is frozen (Stage 3).
5. **Metrics as defined in D4.** Never report accuracy alone.
6. **No silent changes** to data, splits, preprocessing, thresholds, metrics or scope. Propose, explain the impact, decide. Dataset pins change only through a new decision (D21).
7. **Never commit** audio, datasets, checkpoints or secrets. Secrets live in `.env` only.

## 4. How we work

### 4.1 Git workflow

- `main` is protected: changes arrive only through a pull request with **1 approval** and **green CI**.

- **One meaningful unit = one branch = one PR.** Many small commits on a branch are fine; the PR is what gets reviewed.
- Branch names `<type>/stage<N>-<name>` with types `feat`, `fix`, `docs`, `chore` (D23), e.g. `docs/stage3-protocol`, `feat/stage6-preprocessing`. Commit messages follow Conventional Commits (`feat:`, `fix:`, `docs:`, `chore:`).
- **Daily flow:** `git checkout main → git pull → git checkout -b <branch>` → work and commit → push → open PR (base `main`) → review → merge → delete branch.
- **Stacked branches** (when a PR waits for review): the next branch may start from the waiting one; **before merging a stacked PR, switch its base to `main`.**
- Full rules: `CONTRIBUTING.md`. The PR template asks: tests, docs, decision logged, **DEEP-VOICE untouched**, and **which AI tool was used, for what**.

### 4.2 AI tools and human in the loop

Each member uses their own AI coding tool (e.g. Claude Code, Codex, Antigravity with Gemini). All of them read the same rulebook, `AGENTS.md` (D22): Codex and Antigravity natively, Claude Code through `CLAUDE.md`, Gemini CLI through `.gemini/settings.json`.

- **Every AI-produced artifact is reviewed by a team member before it is accepted.** AI tools never merge.
- **Rules are advice; code and CI are the guarantee.** The most important rules are also enforced by tests (e.g. the DEEP-VOICE governance test).
- **Ground truth over self-report:** a claim counts only with real evidence (test output, data, source). Labels: *[verified]*, *[doc]*, *[hypothesis]*.
- AI assistance is recorded honestly in the project log and in each PR.
- **Efficient use:** one fresh AI session per task; prerequisites done before prompting; exact file paths instead of "search for"; short reports.

### 4.3 Documentation system

| Document | Purpose |
|---|---|
| `docs/decisions/README.md` | Register of every decision (D-IDs) and pre-registered hypotheses. Major decisions also have an ADR in `docs/decisions/` (D24). |
| `docs/project_log.md` | The project's story: what happened, what was found, why. Append-only. Its **Current status** section is the first thing to read. |
| `docs/CHANGELOG.md` | What changed in the product (from Stage 6 on). |
| `docs/dataset_audit.md`, `docs/threat_model.md`, `docs/experimental_protocol.md` | Stage deliverables, each with a team sign-off table. |
| `docs/architecture.md`, `docs/setup.md`, `docs/README.md` | Living summary of the SDD, how to run everything, index of all docs. |
| `docs/results/` | Results write-ups per experiment (Stages 7, 11, 13–16). |

### 4.4 Data and compute

- Audio never goes on personal machines or in Git. Datasets live in the shared Google Drive (Appendix C), as **one archive per dataset** (fast to copy), and are copied to Colab's local disk at the start of each session.
- Both datasets are **pinned** (D21): FoR by its Hugging Face revision, DEEP-VOICE by its zip SHA-256. The notebooks refuse to run on unpinned or changed data.
- **Compute:** Google Colab first; Kaggle Notebooks as fallback (≈30 GPU-hours/week per account). Whether a paid tier is needed is decided from the Stage 6 compute benchmark, not guessed.
- All code takes data paths from config files or arguments; nothing is hard-coded to Colab or Drive.

### 4.5 Reproducibility

- Every result = **pinned dataset + config file + code commit**. Configs live in `configs/*.yaml`; every training run is logged to Weights & Biases with its config.
- Python 3.13 is the official version (Colab runtime, D15); 3.14 allowed locally. `requirements.txt` pins exact versions; CI tests on 3.13.
- Fixed random seeds; the final CNN configuration is trained with 3 seeds (D11).

### 4.6 What a stage gate means

A stage is closed when: (1) its outputs are merged into `main` through a reviewed PR with green CI; (2) its sign-off or review is recorded; (3) the project log has an entry and an updated Current status. Only then does the next stage depend on it.

## 5. Team, ownership and reviews

Each part of the plan has one **owner**: the person who drives it, opens its PRs and answers for its quality. Owning a part does not mean working alone: every PR is reviewed by a teammate, and everyone must be able to explain the whole pipeline.

| Member | Owns | Main start | Can start earlier with |
|---|---|---|---|
| **Donia Mabrouk** | Stage 3 (lead) · **Part C**: Stages 7–12 | Stage 3 now; Part C after Stage 6 | The evaluation harness (Stage 7 metrics module), tested on synthetic scores while Part B runs |
| **Malak Ben Salem** | **Part B**: Stages 4–6 · **Stage 14** (robustness) | After Stage 3 | The degradation generator for Stage 14 (noise, codecs, band-limiting) right after the preprocessing profile; it reuses the same signal-processing code |
| **Khalil** | **Stages 13, 15, 16** (external run, calibration analysis, error analysis) | Stage 13 after Stage 12 | Onboarding (Appendix E); his Stage 3 sections; all analysis tools (per-speaker tables, clustered bootstrap CIs, reliability diagrams, error reports) built and tested on synthetic scores or FoR validation outputs |
| **Iheb Zemzemi** | **Part E**: Stages 17–19 (API, dashboard, integration & delivery) | After the Stage 3 interfaces are fixed | API and dashboard against a **mock model** that follows the agreed contract; Docker and integration tests |
| **Everyone** | Stage 3 sign-off · Stage 13 review · Stage 19 report and presentation | — | Each person writes the report section for their own stages as soon as each stage closes |

> **Separation that protects the result.** The DEEP-VOICE external run (Stage 13) is executed by Khalil, who does not build the model (Part C, Donia). The person who trained the model never touches the external test, so nobody is tempted to "adjust one thing" after seeing it.

### Interfaces between parts (fixed in Stage 3)

Different owners work in parallel only if the hand-overs are written down first. These contracts are part of Protocol v1.0; changing them later needs a decision.

| Hand-over | What is fixed in advance |
|---|---|
| Part B → Part C<br>Malak → Donia | Manifest columns (path, label, subgroup, group, split, hash); the preprocessing function's input and output (sample rate, length, feature shapes). |
| Part C → Part D<br>Donia → Khalil | ModelBundle format and a single scoring function, `score(audio) → score`; the per-sample output file format for stored scores. |
| Part C → Part E<br>Donia → Iheb | API response `{prediction, confidence, risk_level}` plus model/profile/policy versions; how the bundle is loaded and its hash verified. |
| Part B → Stage 14<br>Malak | Degradation conditions and levels (from the protocol), applied through the same preprocessing profile. |

### Who reviews what: the consumer reviews the producer

The person who will *use* an output reviews it, which also tests the hand-over. The author never approves their own PR.

| Work | Reviewer(s) |
|---|---|
| Part B (Malak) | Donia — she builds on the manifests and the preprocessing |
| Part C (Donia) | Khalil and Iheb — they use the frozen bundle |
| Stages 14 (Malak), 15–16 (Khalil) | Donia — review only; the frozen model is never changed |
| Part E (Iheb) | Malak |
| **Stage 3 protocol and Stage 13 external run** | **All four** |

The distribution is recorded as decision D26 in the first Stage 3 pull request, together with per-folder entries in `CODEOWNERS` (for example `src/preprocessing/` → Malak, `api/` → Iheb).

## 6. Roadmap at a glance

Target weeks refer to a 14-week semester and are guides, not deadlines; the team aligns them with the course calendar.

| # | Stage | Owner | Target | Status | Gate (what "done" means) |
|---|---|---|---|---|---|
| 0 | Repository & environment | Donia | Wk 1 | DONE | Repo, protection, Python version, datasets pinned |
| 1 | Dataset audit | Donia | Wk 1–2 | DONE | Audit signed off (PR #2) |
| 2 | Threat model | Donia | Wk 2 | DONE | v1.0 reviewed (PRs #3, #4) |
| — | Repository structure (CS 488 Lab 6) | Donia | Wk 2 | DONE | PR #6 merged; CI required on `main` |
| 3 | Experimental Protocol v1.0 | Donia (lead) + all | Wk 3 | NEXT | Protocol frozen (F1), signed by all four |
| 4 | Leakage-aware split | Malak | Wk 3 | TO DO | Split built and checked |
| 5 | Dataset freeze | Malak | Wk 3 | TO DO | Manifests + hashes committed (freeze F2) |
| 6 | Preprocessing profile v1 + compute benchmark | Malak | Wk 4 | TO DO | Profile frozen and tested (F3) |
| 7 | Evaluation harness + MFCC/SVM baseline | Donia | Wk 4–5 | TO DO | Harness verified; baseline results |
| 8 | CNN training | Donia | Wk 5–7 | TO DO | Stable training, all runs logged |
| 9 | Model selection (FoR val-A) | Donia | Wk 7 | TO DO | Final configuration chosen |
| 10 | Risk policy + calibration fit (FoR val-B) | Donia | Wk 8 | TO DO | Thresholds and calibration documented |
| 11 | FoR internal test (once) | Donia | Wk 8 | TO DO | In-domain results recorded |
| 12 | **ModelBundle freeze** | Donia | Wk 8 | TO DO | Bundle immutable (F4) |
| 13 | **DEEP-VOICE external run (once)** | Khalil | Wk 9 | TO DO | Run registered; per-sample scores stored |
| 14 | Robustness (FoR test only) | Malak | Wk 10 | TO DO | Degradation curves |
| 15 | Calibration / reliability analysis | Khalil | Wk 10 | TO DO | ECE and reliability diagrams |
| 16 | Error analysis | Khalil | Wk 11 | TO DO | Errors explained by subgroup, speaker, confidence |
| 17 | API (FastAPI) + Docker | Iheb | Wk 6–11 | TO DO | Integration tests green |
| 18 | Dashboard (Streamlit) | Iheb | Wk 7–12 | TO DO | Usable by a non-technical person |
| 19 | Testing, documentation & delivery | Iheb + all | Wk 13–14 | TO DO | Report, demo, presentation delivered |

**Freeze points**, in time order — F1 protocol (Stage 3) → F2 dataset split + manifests (Stage 5) → F3 preprocessing profile (Stage 6) → F4 ModelBundle (Stage 12). After a freeze, the frozen item changes only through a new decision.

## 7. Part A — Foundation (Stages 0–3) · Donia

### 0 Repository & environment — DONE

**RESULT** GitHub repo with protected `main` (PR + 1 approval); Python 3.13 official (D15); both datasets downloaded, backed up in Drive and pinned (D21).

### 1 Dataset audit — DONE

**RESULT** Full audit of FoR-2sec and a metadata-only audit of DEEP-VOICE, with rules written **before** seeing results. Signed off via PR #2. Decisions D5, D9, D13, D14, D16–D20; hypotheses H5 (MP3 traces) and H6 (bandwidth). Evidence in `docs/audit_evidence/`.

### 2 Threat model — DONE

**RESULT** `docs/threat_model.md` v1.0: bank call-center triage of high-risk requests (main scenario), executive voice message (secondary), threats T1–T7 mapped to the experiment that tests each, system threats, residual risks (incl. oracle abuse). Reviewed via PRs #3 and #4.

### — Repository structure (CS 488 Lab 6 template) — DONE

**CONTENT** `AGENTS.md` as single AI rulebook; `CONTRIBUTING.md`, PR and issue templates, `CODEOWNERS`; decision register + ADRs; `.gitattributes`, `.editorconfig`, `.env.example`; CI on Python 3.13; DEEP-VOICE governance test; docs (architecture, setup, changelog, setup report); MIT license (code only, D25).

**RESULT** PR #6 merged; “Require status checks → CI” enabled on `main`.

**REMAINING** Each member tests their own AI tool on the project rules; Lab 6 setup report filled in (small docs PR).

### 3 Experimental Protocol v1.0

Owner: Donia (lead) NEXT

**GOAL**

Fix every rule of the experiment **before** any model exists, so results cannot shape the rules that judge them.

**TASKS**

- **Research questions:** RQ1 FoR → DEEP-VOICE generalization; RQ2 CNN vs baseline in-domain and externally; RQ3 robustness under degradation; RQ4 calibration under shift.
- **Hypotheses:** H1–H4 (CNN beats baseline in-domain; external EER higher than in-domain; calibration worsens externally; EER rises with degradation) to be confirmed; H5–H6 already registered.
- **Split procedure** (D13, D14, D16, D9): group-aware 70/15/15, stratified by REAL / FAKE_mp3 / FAKE_other; validation halved into val-A (training decisions) and val-B (thresholds, calibration).
- **Metric definitions** (D4) and statistics (D11): 3 seeds, bootstrap 95% CIs, clustered by speaker for DEEP-VOICE.
- **Decision policy** (D12): operating threshold at the FoR val-B EER point; uncertain band and maximum deferral rate.
- **External-run procedure** (D5, D17–D19): 2-s windows; speech-window filter with its threshold fixed here; filtered + unfiltered views; per-speaker table; what is stored.
- **Robustness conditions** (FR-17): noise, compression, bandwidth reduction (incl. ≈3.4 kHz telephone band for H6), with levels.
- **Interfaces between parts** (Section 5): manifests and preprocessing I/O, ModelBundle `score()` and stored-score format, API response contract.
- Freeze points F1–F4 and what each freezes.

**AUTHORS**

Each owner writes the section they will later execute; Donia integrates. **Malak:** split, preprocessing, robustness conditions. **Donia:** metrics, baseline/CNN plan, risk policy and calibration fit. **Khalil:** external-run procedure, calibration and error-analysis plans. **Iheb:** interfaces (bundle scoring function, API contract).

**OUTPUT**

`docs/experimental_protocol.md` v1.0 with a sign-off table.

**GATE**

Protocol merged and signed by all four (freeze F1). **No training code before this gate.**

## 8. Part B — Data & preprocessing (Stages 4–6) · Malak

### 4 Leakage-aware split

Owner: Malak

**TASKS**

- Build the FoR split exactly as the protocol says: near-duplicate groups (threshold 0.95) never cross splits; proportions of REAL / FAKE_mp3 / FAKE_other kept in every split; val-A / val-B.
- Automated checks: no group in two splits, class and subgroup proportions, split sizes.

**OUTPUT**

Split code in `src/` with tests; split report.

**GATE**

All checks pass and are reviewed.

### 5 Dataset freeze (F2)

Owner: Malak

**TASKS**

Write manifests (path, label, subgroup, group, split, hash) for FoR and a DEEP-VOICE file manifest (metadata only, D20); record their hashes.

**OUTPUT**

`data/manifests/*.csv` + hashes committed; decision entry recording the freeze.

**GATE**

Manifests merged; the split can no longer change without a new decision.

### 6 Preprocessing profile v1 + compute benchmark (F3)

Owner: Malak

**TASKS**

- One shared pipeline for training, testing and the API: channel averaging to mono and resampling to 16 kHz (D18); silence handling; fixed 2-s input length; MFCC features (baseline) and log-Mel spectrograms (CNN).
- **Decide the silence detector explicitly** (energy-based vs. a small VAD model such as silero-vad) and record it as a decision; the D17 filter uses the same detector.
- Unit test every function. **Pipeline-invariance test**: FoR clips converted to 40 / 44.1 / 48 kHz stereo must give near-identical features (D18).
- **Re-run the shortcut screen after preprocessing** (silence, bandwidth; D16).
- **Compute benchmark:** train the CNN briefly on a small subset; record GPU, VRAM, batch size, time per epoch → decide Colab free / Pro / Kaggle.
- Create the W&B project and shared logging setup.

**OUTPUT** `src/preprocessing/` + tests; versioned profile config; benchmark note; changelog starts.

**GATE** Profile frozen (F3); invariance and shortcut checks reviewed.

## 9. Part C — Modeling & freeze (Stages 7–12) · Donia

### 7 Evaluation harness + MFCC/SVM baseline

Owner: Donia

**TASKS**

- One reusable metrics module implementing D4 (EER, ROC-AUC, decision metrics, 3-way confusion matrix, coverage, CIs), tested against known values.
- Baseline: MFCC mean + std → RBF-SVM (D7), trained on FoR train, tuned on val-A.
- Log to W&B with config; write the baseline results while fresh.

**OUTPUT** `src/evaluation/`, `src/models/baseline.py`, `docs/results/baseline.md`.

**GATE** Harness verified; baseline results reviewed.

### 8 CNN training

Owner: Donia

**TASKS** One CNN on log-Mel spectrograms, trained on FoR train only; no augmentation (D10); every run logged to W&B, including “quick tests”. If results look too good, suspect leakage or shortcuts first.

**OUTPUT** `src/models/` CNN + training code; `configs/cnn.yaml`.

**GATE** Stable, reproducible training.

### 9 Model selection (FoR val-A only)

Owner: Donia

**TASKS** Compare configurations on val-A; choose the final one; train it with 3 seeds (D11). The test split and DEEP-VOICE stay untouched.

**GATE** Final configuration documented with its evidence.

### 10 Risk policy + calibration fit (FoR val-B)

Owner: Donia

**TASKS** Fit calibration (temperature or Platt scaling) and the decision policy on val-B (D2, D9, D12): operating threshold, uncertain band, LOW / MEDIUM / HIGH mapping. Implement as a standalone, tested function: raw score → {prediction, confidence, risk level}.

**OUTPUT** `src/risk/` + tests; documented threshold values and reasoning.

### 11 FoR internal test (once)

Owner: Donia

**TASKS** Evaluate baseline and CNN on the FoR test split, once. Report all D4 metrics, per subgroup (H5: FAKE_mp3 vs FAKE_other).

**OUTPUT** `docs/results/in_domain.md`.

### 12 ModelBundle freeze (F4)

Owner: Donia

**TASKS** Package checkpoint + preprocessing profile + risk policy as one versioned bundle with hashes; record the freeze as a decision.

**GATE** **Bundle immutable.** From here on nothing is tuned.

## 10. Part D — External test & analysis (Stages 13–16) · Khalil, Malak

### 13 DEEP-VOICE external run — exactly once

Owner: Khalil

**TASKS**

- One guarded run of the frozen bundle, following the protocol: mono + 16 kHz (D18), 2-s windows (D5), speech-window filter with filtered and unfiltered views from the same run (D17).
- Store **per-sample scores**, so every later analysis reads stored outputs and never re-runs the model.
- Report all D4 metrics with the 87.5% FAKE imbalance stated, CIs clustered by speaker, a per-speaker table and a speaker average (D19).
- State the scope: generalization to RVC voice conversion, not to all voice cloning (D8).

**OUTPUT** `docs/results/generalization.md` — the project's central finding.

**GATE** Run registered once; results reviewed by all four. A large drop is a valid, reportable finding, not a failure.

### 14 Robustness (FoR test only, D3)

Owner: Malak

**TASKS** Apply the protocol's degradations (noise, compression, bandwidth reduction) at increasing levels and plot EER per level. Telephone-band REAL clips test H6: genuine callers must not be flagged as synthetic.

**OUTPUT** `configs/robustness.yaml`, `docs/results/robustness.md`.

**START** The degradation generator can be built right after Stage 6; the curves are produced once the bundle is frozen (Stage 12).

### 15 Calibration / reliability analysis

Owner: Khalil

**TASKS** Reliability diagrams and ECE on FoR test and on the **stored** DEEP-VOICE scores (no re-run). Does the “uncertain” band still mean something under shift?

### 16 Error analysis

Owner: Khalil

**TASKS** Where and why the model fails: by subgroup, speaker, confidence level and deferral; link findings back to the threat model (threats T1–T7).

**OUTPUT** `docs/results/error_analysis.md`.

## 11. Part E — System & delivery (Stages 17–19) · Iheb

### 17 API (FastAPI) + Docker

Owner: Iheb

**TASKS**

- `POST /analyze-audio`: same preprocessing → frozen bundle → risk layer → JSON {prediction, confidence, risk_level}.
- SRS requirements: upload size / duration limits, sanitized errors, logging without raw audio, request ID and versions per record, retention limit.
- Threat-model controls: verify the bundle hash at load; decide access control (no public API, authenticated clients, rate limits) to address oracle abuse.
- Integration tests; Dockerfile and `docker-compose.yml`.

**START** Scaffolding against a mock model can start during Stage 8; the real bundle is plugged in after Stage 12.

**GATE** `docker-compose up` serves a working API; integration tests green in CI.

### 18 Dashboard (Streamlit)

Owner: Iheb

**TASKS** Submit audio; show prediction, confidence and risk level; history of submissions; analyst notes if FR-11 is implemented. Wording is decision-support ("likely synthetic — verify"), never a verdict.

**START** Can start against mocked API responses during Stage 8.

**GATE** A non-technical person understands a result without explanation; containerized.

### 19 Testing, documentation & delivery

Owner: Iheb + all

**TASKS**

- Full regression pass; CI green; bug triage.
- Architecture doc updated to what was actually built; README stands alone as a portfolio piece; setup report complete.
- Final technical report: requirements, threat model, protocol, results, limitations, and an honest section on how AI tools were used and verified.
- Demo audio set mixing in-domain and out-of-domain examples (DEEP-VOICE clips only after the external run is registered).
- Rehearse the presentation end to end at least twice. Optional: public demo on Hugging Face Spaces.

**GATE** Report submitted; demo and presentation delivered.

### Parking lot (only if both stretch experiments are done with real time to spare)

- Generator- or speaker-specific held-out evaluation within FoR (not possible with the current metadata).
- A second external dataset (ASVspoof, WaveFake, In-the-Wild).
- wav2vec2 / WavLM fine-tuning; spectrogram explainability (lowest priority).

## 12. Work before your own stages start

The ML stages are sequential by design, so each owner has preparatory work that does not depend on the previous gate. Nobody should wait idle.

| Owner | When | Preparatory work |
|---|---|---|
| Donia | While Part B runs | Evaluation harness (D4 metrics, CIs) tested on synthetic scores. No training code before Stage 3 is frozen. |
| Malak | After Stage 6 | Stage 14 degradation generator, tested on FoR samples (no model needed). |

| | | |
|---|---|---|
| Khalil | From now | Onboarding; Stage 3 sections; analysis tools for Stages 13, 15 and 16 on fake or FoR-validation scores; on the day of Stage 13 he only runs them, once. |
| Iheb | After Stage 3 | API and dashboard against a mock model with the agreed contract; swap in the frozen bundle after Stage 12. |
| Everyone | From Stage 3 | Report sections written when each stage closes; GitHub Issues board (one issue per SRS requirement, one milestone per stage). |

Everyone should be able to explain the full pipeline end to end, not only the part they built.

## 13. Risks and mitigations

| Risk | Mitigation |
|---|---|
| DEEP-VOICE accidentally used for tuning | Hard rule 1; D20 permitted accesses; CI governance test; single guarded run after the freeze. |
| Model learns dataset artifacts (MP3 history, bandwidth) instead of synthesis | Subgroup-stratified split (D16); pre-registered H5 and H6; shortcut screen re-run after preprocessing. |
| Genuine phone callers flagged as synthetic | Telephone-band robustness test (H6); “uncertain” band; decision-support wording. |
| Leakage between splits (no speaker metadata) | Near-duplicate groups kept in one split (D14); in-domain results reported as an upper bound. |
| Our own pipeline creates a domain shift | Identical preprocessing everywhere; pipeline-invariance test (D18). |
| Over-claiming generalization | Scope stated as RVC only (D8); 8 speakers → wide CIs reported; per-speaker table. |
| Insufficient GPU | Compute benchmark in Stage 6; Kaggle fallback; lightweight CNN by design. |
| Scope creep | Scope lock; parking lot only after both stretch experiments. |
| AI-generated errors accepted unchecked | Human in the loop; real test output required; reviews on every PR. |
| Team fragmentation | Weekly sync; every PR reviewed by a teammate; shared project log. |
| Hand-over gaps between owners | Interfaces fixed in Stage 3; the consumer reviews the producer (Section 5). |
| New member joins mid-project | Structured onboarding (Appendix E); first PR is small; he co-authors the protocol he will execute. |

## 14. Open items before Stage 3

1. **Khalil's onboarding** (Appendix E): GitHub access, setup, first small PR.
2. Each member tests their own AI tool in a fresh session (does it load `AGENTS.md`?); answers go into the Lab 6 setup report, then one small docs PR closes the structure step.
3. Team meeting: weekly sync time and channel; compute access per person; Weights & Biases project.
4. Record the distribution (D26), team names in `README`, `AGENTS.md` and `LICENSE`, and per-folder `CODEOWNERS`, in the first Stage 3 PR.
5. Create the GitHub Issues board (one issue per SRS requirement, milestones per stage).
6. Confirm with the professor that `AGENTS.md` as the single AI context (with `AI_CONTEXT.md` as a pointer) is acceptable for Lab 6.
7. Housekeeping: regenerate the Kaggle API token; answer the three open questions in the threat model (§13).

## Appendix A — Decision register summary

Authoritative text: `docs/decisions/README.md`. This table is a summary.

| ID | Decision |
|---|---|
| D1 | Write an Experimental Protocol after the audit; freeze it before training. |
| D2 | Fit risk policy and calibration on FoR validation **before** the ModelBundle freeze; the DEEP-VOICE run comes after. |
| D3 | No robustness evaluation on DEEP-VOICE; robustness on the FoR test split only. |
| D4 | Metric definitions: score-level (EER, ROC-AUC) and decision-level metrics, 3-way confusion matrix, coverage, CIs; positive class = spoof. |
| D5 | Evaluation unit: 2-s windows (primary), file-level mean score (secondary). Confirmed by the audit. |
| D6 | DEEP-VOICE audit is metadata-only. |
| D7 | Baseline = RBF-SVM on MFCC mean + std. |
| D8 | Research question narrowed to FoR (TTS) → DEEP-VOICE (RVC) generalization. |
| D9 | Validation split into val-A (training decisions) and val-B (thresholds, calibration). |
| D10 | No degradation augmentation in model v1. |
| D11 | 3 seeds on the final CNN configuration; bootstrap 95% CIs, clustered by speaker for DEEP-VOICE. |
| D12 | Operating threshold at the FoR validation EER point, plus a fixed deferral policy. |
| D13 | FoR is re-split (no packaged split exists). |
| D14 | Grouping by near-duplicate components at similarity 0.95 (no speaker metadata). |
| D15 | Official Python 3.13 (Colab); 3.14 allowed locally; CI on 3.13. |
| D16 | MP3 history: no data changes; split stratified by REAL / FAKE_mp3 / FAKE_other; silence re-checked after preprocessing. |
| D17 | DEEP-VOICE speech-window filter (frozen detector, threshold fixed in the protocol); filtered + unfiltered views. |
| D18 | Mono averaging + 16 kHz resampling everywhere; pipeline-invariance test at 40 / 44.1 / 48 kHz. |
| D19 | External results include a per-speaker table and a speaker average. |
| D20 | Permitted DEEP-VOICE accesses before the external run. |
| D21 | Datasets pinned (FoR revision, DEEP-VOICE zip SHA-256); pins change only by a new decision. |
| D22 | `AGENTS.md` is the single AI context for all tools. |
| D23 | Branch naming `<type>/stage<N>-<name>`. |
| D24 | Decision register + ADRs for major decisions only. |
| D25 | MIT license for the code only; datasets never redistributed. |
| H5 | If the CNN relies on MP3 traces, its recall on FAKE_other is clearly lower than on FAKE_mp3. |
| H6 | If the CNN relies on bandwidth, band-limiting REAL clips to ≈3.4 kHz raises false “synthetic” calls. |

## Appendix B — Repository structure

```
voice-spoofing-detection/
├── AGENTS.md                 single AI rulebook (all tools)
├── CLAUDE.md · AI_CONTEXT.md pointers to AGENTS.md
├── README.md · CONTRIBUTING.md · LICENSE
├── requirements.txt · pytest.ini
├── .gitignore · .gitattributes · .editorconfig · .env.example
├── .gemini/settings.json     Gemini CLI reads AGENTS.md
├── .githooks/pre-push        docs reminder (never blocks)
├── .github/                  CI workflow, PR + issue templates, CODEOWNERS
├── .vscode/extensions.json
├── scripts/                  setup.sh · check-docs.sh · update-structure.sh
├── configs/                  audit.yaml (dataset pins) · baseline.yaml · cnn.yaml · robustness.yaml (Stage 14)
├── data/                     README.md · manifests/ (small CSVs only — no audio)
├── notebooks/                Colab entry points (01_stage1_dataset_audit.ipynb …)
├── src/
│   ├── audit/                Stage 1 tools (only code allowed to touch DEEP-VOICE metadata)
│   ├── preprocessing/        Stage 6
│   ├── models/               Stages 7–9
│   ├── evaluation/           Stage 7
│   └── risk/                 Stage 10
├── tests/                    mirrors src/ · governance/ (DEEP-VOICE access guard)
├── api/                      Stage 17 (FastAPI) · docker-compose.yml (Stage 17)
├── dashboard/                Stage 18 (Streamlit)
└── docs/
    ├── README.md · architecture.md · setup.md · setup-report.md · CHANGELOG.md
    ├── project_log.md        story, open items, current status
    ├── decisions/            register (README.md) + ADRs
    ├── dataset_audit.md · threat_model.md · experimental_protocol.md (Stage 3)
    ├── results/              Stage 7+ write-ups
    ├── audit_evidence/       committed audit outputs
    └── reference/            Proposal, SRS, SDD, Execution Plan
```

## Appendix C — Shared Drive structure

```
MyDrive/voice-spoofing-detection/
├── raw/
│   ├── for_2sec_ff8c82c7.tar                    FoR-2sec archive (1.22 GB) + for_2sec_REVISION.txt
│   └── deep-voice-deepfake-voice-recognition.zip DEEP-VOICE (3.96 GB) + deep_voice_ZIP_SHA256.txt
├── processed/
│   ├── metadata/stage1/      full audit outputs (inventories, fingerprints, summaries)
│   ├── manifests/            mirror of data/manifests (Stage 5)
│   └── features/             cached features (Stage 6+)
└── checkpoints/              baseline/ · cnn/ · experiments/<W&B run ID>/ · bundle/ (Stage 12)
```

Storage: ≈5.2 GB in total, within a free 15 GB Google account. Copy archives to Colab's local disk before heavy reading; reading thousands of small files directly from Drive is slow.

## Appendix D — What changed since v1

| v1 said | v2.0 does |
|---|---|
| DEEP-VOICE tested in Week 8, risk layer in Week 9, calibration in Week 12 | Thresholds and calibration fitted on FoR **before** the freeze; DEEP-VOICE after (D2). The v1 order would have broken its own freeze rule. |
| Robustness “optionally” on DEEP-VOICE | Robustness on FoR only; DEEP-VOICE is touched once (D3). |
| “Explore both datasets” | DEEP-VOICE inspected by metadata only (D6, D20). |
| No experimental protocol document | Stage 3 protocol frozen before any training (D1). |
| GMM or SVM baseline | RBF-SVM on MFCC mean + std (D7). |
| Augmentation with noise and codecs | No augmentation in v1, so robustness results stay clean (D10). |

| | |
|---|---|
| DEEP-VOICE: 628 bonafide / 4,425 spoof, ≈0.55 GB | Verified: 8 REAL / 56 FAKE files, 1,870 / 13,090 windows, 3.96 GB. |
| FoR spans six named TTS engines | Not determinable from the package; recorded as a limitation. |
| Group split “if speaker metadata exists” | No speaker metadata; near-duplicate groups (D14) + subgroup stratification (D16). |
| Data downloaded straight into Drive folders | One archive per dataset in Drive, pinned and hash-checked (D21). |
| 11 phases over 14 weeks | 20 stages with explicit gates and freeze points; same deliverables. |
| Generic repository layout | CS 488 Lab 6 structure adapted to the project; one AI rulebook for all tools (D22–D25). |
| Three members, tracks by role | Four members (v2.1); ownership by plan part, interfaces fixed in Stage 3, consumer-reviews-producer rule (D26). |

## Appendix E — Onboarding a new member

Goal: a new member understands **why** the rules exist, can run the repository, and has merged a first PR within a few days. Buddy for Khalil: **Donia**.

### Day 1 — Access and setup (≈2 hours)

1. Accept the GitHub invitation; he is added to `CODEOWNERS`.
2. Get access to the shared Drive folder and the team channel; Weights & Biases when it exists.
3. Clone, create the virtual environment (Python 3.13, or 3.14 locally), `pip install -r requirements.txt`, run `pytest -q` (expected: all tests pass), run `bash scripts/setup.sh` (Git Bash on Windows). Guide: `docs/setup.md`.
4. AI tool test: in a fresh session in the repository, ask *"What are this project's hard rules about DEEP-VOICE, and where would preprocessing code go?"*. The answer must mention "touched exactly once / D20" and `src/preprocessing/`.

### Day 1–2 — Understand the project (reading order, ≈3 hours)

| Read | To understand |
|---|---|
| This plan, Sections 1–6 | Goal, rules, workflow, ownership, roadmap |
| `AGENTS.md` and `CONTRIBUTING.md` | The rules every person and AI tool follows; branches, PRs, reviews |
| `docs/project_log.md` — Current status, then skim the entries | What happened so far and why |
| `docs/dataset_audit.md` §7–§8 | The datasets, the decisions taken from the audit, the limitations |
| `docs/threat_model.md` §3 and §8 | The scenario and which experiment tests which threat |
| `docs/decisions/README.md` | All decisions D1–D25; especially D2, D5, D8, D11, D17–D20 for his stages |

### Day 2 — Walkthrough call with the buddy (≈45 minutes)

- Repository and Drive tour; how a stage moves from branch to merged PR.
- The story behind the rules: why DEEP-VOICE is touched once, why thresholds are fitted before the external run, what the MP3 and bandwidth findings mean.
- His stages (13, 15, 16): inputs he will receive, outputs he must produce, who reviews them.

### Days 2–4 — First contribution

1. **First small PR** to learn the workflow: his tool-test answer in `docs/setup-report.md`, or the Issues board milestones for Part D. Branch → PR template → CI → review → merge.
2. **Stage 3 sections**: external-run procedure (D5, D17–D19), calibration-analysis and error-analysis plans.
3. **Understanding check**: a 5-minute explanation to the team of the research question, why DEEP-VOICE is used once, and what his stages produce.

> **Never, from day one:** open, play or plot any DEEP-VOICE file; push to `main`; merge without review; commit data, checkpoints or secrets; let an AI tool merge or change frozen items.

Prepared from the Proposal, SRS v3.0, SDD v1.0, the v1 Execution Reference, the project log and the decision register, with AI assistance (Claude); v2.1 adds team ownership; to be reviewed by the team.